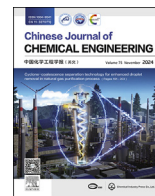




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Steady-state and dynamic simulation of gas phase polyethylene process

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ABSTRACT

Gas-phase polyethylene (PE) processes are among the most important methods for PE production. A deeper understanding of the process characteristics and dynamic behavior, such as properties of PE and reactor stability, holds substantial interest for both academic researchers and industries. In this study, both steady-state and dynamic models for a gas-phase polyethylene process are established as a simulation platform, which can be used to study a variety of operation tasks for commercial solution polyethylene processes, such as new product development, process control and real-time optimization. The copolymerization kinetic parameters are fitted by industrial data. Additionally, a multi-reactor series model is developed to characterize the temperature distribution within the fluidized bed reactor. The accuracy in predicting melt index and density of the polymer, and the dynamic behavior of the developed models are verified by real plant data. Moreover, the dynamic simulation platform is applied to compare four practical control schemes for reactor temperature by a series of simulation experiments, since temperature control is important in industrial production. The results reveal that all four schemes effectively track the setpoint temperature. However, only the demineralized water temperature cascade control demonstrates excellent performance in handling disturbances from both the recycle gas subsystem and the heat exchange subsystem.

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1. Introduction

Polyethylene (PE) is a highly versatile synthetic material widely used in various applications in our daily lives. It is one of the most produced synthetic materials worldwide. Among the different PE production processes, gas-phase processes have gained prominence due to their simplicity, versatility, and environmental advantages, particularly in terms of energy savings [1]. Gas-phase fluidized bed polyethylene processes, such as the Unipol process, Innovene process, and Spherilene process, have become prevalent in the industry. The Unipol process, in particular, is recognized as

one of the fundamental technologies in this field [2]. One notable characteristic of gas-phase PE processes is their sensitivity to reaction temperature. The reactor temperature plays a crucial role in determining catalyst activity and product quality. As the reactor temperature approaches the melting temperature of the PE particles [3], maintaining a stable temperature becomes crucial. Oscillations or fluctuations in the reactor temperature can lead to particle melting, agglomeration, and even de-fluidization, which may ultimately result in reactor shutdown [4]. Thus, reactor temperature control is important and also a challenging task for gas-phase PE processes.

To address the challenge of understanding the dynamic behavior and thermal stability of gas-phase polyethylene (PE) processes, particularly fluidized bed reactors (FBRs), extensive modeling efforts have been undertaken. The ethylene polymerization process in the fluidized bed reactor involves a multi-scale

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coupling of physical and chemical processes, which has been modeled at micro, meso, and macro scales [5]. Choi and Ray [6] proposed a two-phase model to describe the FBR, considering two phases: the bubble phase and the emulsion phase, with the reaction occurring in the emulsion phase. This two-phase model has been extensively studied and further improved, leading to various modified versions [7,8], three-phase models [9] and four-phase models [10], aiming to better understand and model the complex mass and energy transport phenomena.

Concerning the reactor behavior, early assumptions considered the FBR to be a well-mixed continuous stirred tank reactor (CSTR) [11], which is an oversimplification given the non-ideal nature of FBRs. Alizadeh *et al.* [12] used multiple series of fully mixed-flow reactors to simulate the non-ideal reactor behavior of FBRs. Jafari *et al.* [13] and Kiashemshaki *et al.* [14] used the multi-reactor series model to describe the FBR, using multiple CSTRs to describe the emulsified phase and multiple PFRs to describe the bubble phase. The volume of the reactor in each stage is assumed the same. Zhou *et al.* [15] divided the reactor into two regions: gas-liquid-solid and gas-solid. The emulsified phase is considered to be plug flow in the gas-liquid-solid region, while the gas-solid region is considered to be fully mixed flow. The bubble phase is considered to be plug flow throughout the reactor. The temperature prediction results of the model are consistent with the industrial measurement data. Recently, Feng *et al.* [16,17] developed a process model, based on a non-standard polymerization reaction scheme from a pilot scale gas phase polymerization process catalyzed by metallocene, which can depict well the activation of dormant ethylene species.

The FBR models have proven to be valuable tools in studying the thermal stability of reactors [18]. Fan *et al.* [1] established a mathematical model of the condensed-mode gas-phase PE process. The results demonstrated that the thermal stability of the FBR is mainly affected by the catalyst activation energy, superficial gas velocity and condensed-liquid flow rate. Recently, they investigate the dynamic operation mode of ethylene polymerization process with a liquid-containing fluidized bed and showed that it is an effective approach to regulate the chain branching distribution of polyethylene [19]. Salau *et al.* [20] developed a mathematical model of the Unipol process. Their findings revealed that when the reactor operated without a feedback temperature controller and the manipulated variable (valve position) reached saturation, it could lead to oscillatory behavior and limit cycles. Additionally, the presence of recycles in the process can introduce integrating effects, potentially leading to a snowball effect [21]. Alves *et al.* [22] established a fluidized bed reactor model including multi-reactor series model, thermodynamics, kinetics, population conservation model and liquid evaporation model, and investigated the effect of coagulant on reactor temperature and yield.

Many researchers have chosen to use process simulation software, such as Aspen Plus, rather than constructing rigorous mathematical models from scratch. Aspen Plus offers the advantage of user-friendly reactor models and thermodynamic models, like the Sanchez-Lacombe (SL) model [23] and the perturbed chain statistical associating fluid theory (PC-SAFT) equation [24]. Additionally, the software allows for easier construction of complete polyethylene processes, serving as a simulation platform. This approach enables the investigation of the effects of operating variables on polymer properties and process stability from a holistic and systematic perspective. Aspen Plus has been widely used in both steady-state and dynamic simulations of various polyethylene processes, such as slurry high density polyethylene process [25], the HYPOL polypropylene process [26], Spheripol polypropylene process [27], solution polyethylene process [28], and gas-phase polypropylene process [29]. Khare *et al.* [29] established a steady-state and dynamic model of a gas-phase polypropylene process,

in which a fully mixed-flow reactor model was used as a fluidized bed reactor, and the dynamic model was applied to the comparison of brand switching control strategies. Shamiria *et al.* [30] developed a steady-state model for a gas-phase polypropylene reactor, utilizing multiple series of fully mixed flow reactors to represent the fluidized bed reactor. The model accurately predicted key variables, such as yield. Shamiria *et al.* [30] also compared the two-phase model and the well-mixed model based on simulation software. The results showed that both models performed well in predicting yield, but the two-phase model outperformed the well-mixed model in predicting weight-average molecular weight, polymer dispersity index (PDI), and melt flow index (MFI) due to the consideration of hydrodynamic correlations. It is recommended that adding hydrodynamic factors to the modeling of simulation software would make the simulation results more accurate.

In industrial gas-phase polyethylene (PE) processes, the control of reactor temperature is closely related to the fouling prevention heat exchange schemes, which typically include a shell and tube heat exchanger and a plate heat exchanger, where bypass valves are introduced. Bypass control has been shown to offer several advantages in temperature control. Luyben [31] demonstrated that bypass control can lead to fast dynamics when blending a cold stream and a hot stream, allowing for tight temperature control in the heat exchange process. Meanwhile, Soave and Barolo [32] demonstrated that bypass control can enhance the rangeability of the heat exchanger, allowing it to operate effectively across a wider range of temperature conditions. However, it is essential to acknowledge that introducing bypass control also brings about increased complexity and challenges in the design of the overall control system. The bypass control may interact with other control variables and potentially cascade with other control loops in the process. Managing such interactions and optimizing the control strategy to achieve the desired performance becomes an important task in control system design, which has not been evaluated yet and it is one of the objectives of this paper.

As previously mentioned, various advanced models have been developed for fluidized bed reactors in gas-phase polyethylene processes. Mathematical models focus on accurately describing the complex physical and chemical phenomena occurring inside the fluidized bed. On the other hand, process simulations aim to provide a comprehensive analysis of the overall process, sacrificing some level of accuracy in exchange for a broader perspective. Indeed, there exists a trade-off between accuracy and comprehensiveness when it comes to modeling gas-phase polyethylene processes. Striking the right balance between these two aspects is essential in developing models that are both sufficiently accurate and capable of providing a holistic understanding of the process dynamics. To address the complexity of gas-phase polyethylene processes and achieve effective control, the design of robust control systems is crucial. A simulation platform that can be easily integrated with control scheme design and analysis, while maintaining a relatively high level of accuracy, is highly desirable.

In this paper, a comprehensive study of an industrial gas-phase polyethylene process was conducted using Aspen Plus and Aspen Plus Dynamics software, incorporating strict thermodynamic and physical models. The process comprises three main subsystems: a reaction subsystem, a recycle gas subsystem, and a heat exchange subsystem with fouling prevention measures. A five-cascade CSTR model is utilized to simulate the fluidized bed reactor, which can predict the temperature distribution of the reactor accurately. The simulated results are validated using industry data, which includes recycle gas component analysis and polymer properties. For the dynamic simulation, a control system for the gas-phase PE process is designed. The dynamics of the manipulated variable valve characteristics and the heat exchange system are also verified through

the simulations. Four different temperature control methods via PID controller are proposed. A series of simulation experiments are carried out to evaluate the setpoint tracking capability, disturbances rejection capability, and the coupling effects of other controllers. The optimal temperature control scheme is selected from the results of the simulation experiment, which provides guidelines for the design of the industrial control system.

2. Overview of Gas-phase PE Process

The fluidized bed reactor (FBR) is a crucial component in the gas-phase polyethylene (PE) process. As shown in Fig. 1, ethylene (monomer, C₂), hydrogen (chain transfer agent, H₂), and butene (comonomer, C₄) are continually introduced into the FBR from the bottom. These compounds react with the catalyst (CAT) and cocatalyst (CCAT) introduced through a catalyst feed valve. Polymerization of ethylene and butene occurs at the active sites of catalyst, and PE grows to particles with diameters of several hundred micrometers. Ethylene polymerization is an exothermic reaction, releasing a significant amount of heat. The reaction heat is managed through the evaporation of the condensing agent isopentane (C₅) and the sensible heat of the reactants and the inert gas nitrogen (N₂). Both isopentane and nitrogen are supplied to the FBR through the same inlet as ethylene and butene. The PE product is discharged from the bottom of the reactor. The single-pass conversion rate of ethylene in the FBR ranges from 2% to 30% [33]. Large amounts of unreacted gas and inert gas will be recycled to the FBR. These gases are mixed with fresh feed, pressurized, and cooled before being reintroduced into the fluidized bed reactor for further polymerization.

Heat removal is of significant importance in the gas-phase PE process due to the temperature sensitivity of the polymerization reaction. To ensure stable and long-term operation while preventing fouling issues, the recycled gas is cooled using demineralized water (DW) instead of conventional cooling water (CW). As shown in Fig. 1, the cooling system for the recycled gas involves a

shell and tube heat exchanger (EX1) and a plate heat exchanger (EX2). The demineralized water absorbs heat from the recycled gas through EX1 and then transfers it to the cooling water through EX2. A portion of the demineralized water bypasses EX2, and the flow rates of the bypass stream and the main stream are controlled by valves V1 and V2. The valve openings of V1 and V2 always add up to 100%, allowing simultaneous control of the flow rates of both streams. This arrangement enables a wider operating range for the cooling system. It should be noted that the composition and phase status of the recycled gas are susceptible to the temperature disturbance. This poses challenges for temperature control in the reactor. An effective control scheme is required to handle these disturbances and maintain stable conditions.

3. Steady-state Simulation of Gas-phase PE Process

The simulation of the gas-phase PE process is conducted in the Aspen Plus. In the following, key simulation settings and results are presented, while more details can be found in the Supporting Information, including the schematic of the flowsheet, etc.

3.1. Thermodynamic model of POLYSL

The pressure of the fluidized bed reactor in the reaction unit is 2.30 MPa. Within the reactor, both hydrocarbon reactants and polymers coexist, and the reactants such as ethylene, butene, and hydrogen are in a supercritical state. The thermodynamic properties and phase equilibrium of the reactants determine the material balance and heat balance of the reactor. An appropriate thermodynamic model is very important for modeling. The Sanchez-Lacombe (SL) physical property method [23] is a molecular theory of classical fluids based on a compressible lattice fluid model. The model assumes that each molecule can be decomposed according to a continuous fragment sequence. The lattice units within the model can either be empty or occupied by molecular fragments corresponding to the components present in the mixture. The

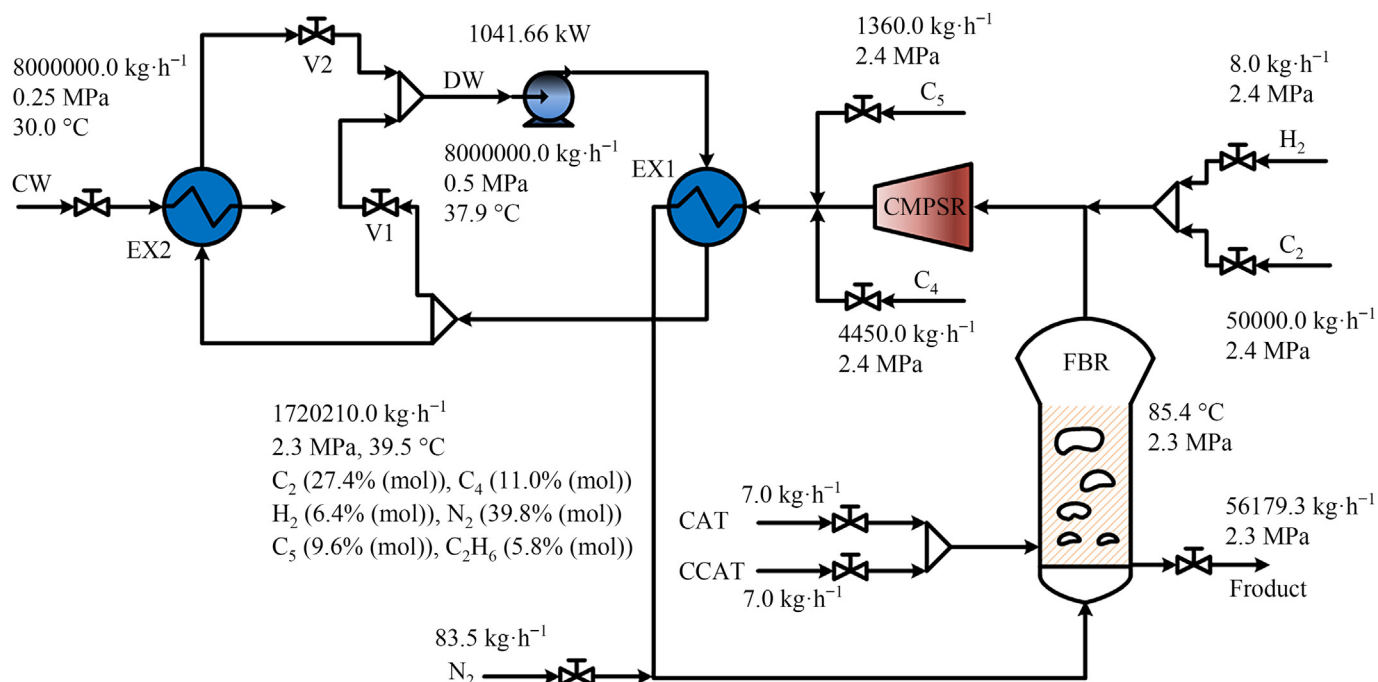


Fig. 1. Flow sheet of an industrial gas-phase PE process.

model has also been extended to the polymer solution system [34]. It is believed that the number of segments of polymer molecules is infinite, and the model can well describe the mixing enthalpy and gas-liquid equilibrium of low molecular weight solutes in polymer liquids. For polymer mixtures, the binary interaction parameters can also be reasonably predicted according to the solubility parameters of the mixture components [35]. As a result, the thermodynamic method known as POLYSL in Aspen Plus is adopted for polymer reaction units. Further details and corresponding pure component and binary component parameters can be found in the Supporting Information.

3.2. Fluidized bed reactor modeling

Many studies, e.g. McAuley et al. [36], use a continuous stirred tank reactor (CSTR) for predicting temperature and concentration in the fluidized bed reactor. However, it is known that the concentration of solid particles in the FBR varies at different positions, and the polymerization reaction occurs in the solid phase. To accurately simulate the temperature distribution within the reactor, fluid mechanics principles are applied. Specifically, the reactor is divided into multiple reaction regions according to the concentration distribution of solid particles, and each region is simulated as a vapor-liquid CSTR. The polymer particles are treated as liquid phase where the polymerization reaction occurs and the mixed gases are treated as vapor phase.

The proportion of solid particles along the height of the fluidized bed reactor can be calculated using the fluidized bed module of Aspen Plus. The model requires the physical properties of the circulating gas, the size design of the reactor and the operating state of the reactor. The minimum fluidization velocity U_{mf} and the minimum fluidization bed porosity ε_{mf} are important process parameters for the fluidized bed fluid calculation. The critical fluidization velocity is calculated as follows [37]:

$$U_{mf} = \frac{Re_{mf} \times \nu_f}{d_p} \quad (1)$$

$$Re_{mf} = (29.5^2 + 0.0357Ar)^{0.5} - 29.5 \quad (2)$$

$$Ar = \frac{g \cdot d_p^3 \cdot (\rho_s - \rho_g)}{\nu_f^2 \cdot \rho_g} \quad (3)$$

The critical bed porosity is calculated as:

$$\varepsilon_{mf} = 0.586 \times \phi^{-0.72} \times \left(\frac{\mu^2}{\rho_g \eta d_p^3} \right)^{0.029} \times \left(\frac{\rho_g}{\rho_c} \right)^{0.021} \quad (4)$$

where ϕ is the particle sphericity, which is 0.776. According to the laboratory determination of polyethylene particle size, polyethylene particle size distribution is shown in Table 1. The reactor equipment parameters and the circulating gas properties are listed in Table 2. According to these parameters, the proportion of solid particles along the height of the fluidized bed reactor can be obtained by the fluidized bed module, as shown in Fig. 2.

In Fig. 2, the solid phase ratio at different heights of the reactor is divided into five distinct regions based on the calculation results. Each region is then simulated using a CSTR, whose volume is determined according to the proportion of particles. The proportion of particles in each region is determined by calculating the average proportion of particles within that specific section, as Eq. (4). The resulting proportions of particles for the five regions are 0.31, 0.32, 0.225, 0.10, and 0.003414, respectively.

Table 1
Polyolefin particle size distribution

Number	Upper limit/mm	lower limit/mm	Mass ratio	Cumulative mass proportion
1	0.0381	0.0965	0.005	0.005
2	0.0965	0.1778	0.0116	0.0166
3	0.1778	0.3531	0.1463	0.1629
4	0.3531	0.7061	0.445	0.6079
5	0.7061	1.4097	0.3808	0.9887
6	1.4097	2.8194	0.0113	1

Table 2
Hydrodynamic calculation parameters

Variable	Value
Height/m	18.0
Diameter/m	5.3
Height of the enlarged section/m	6.75
Diameter of enlarged segment/m	7.32
Temperature of reactor/°C	80
Diameter of solid particles/mm	0.09375
Density of circulating gas/kg·m ⁻³	30
Viscosity of circulating gas/Pa·s	1.315 × 10 ⁻⁵
Density of solid particles/kg·m ⁻³	943.3

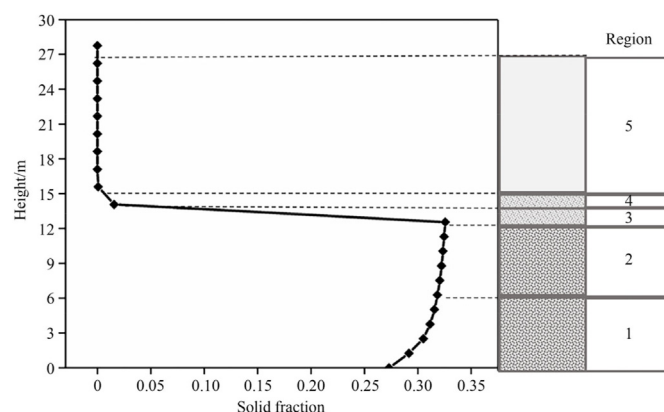


Fig. 2. The proportion of solid phase at different reactor height. Regions 1–5 represent reactor sections with different solid fractions.

$$\bar{\varepsilon}_i = \frac{1}{Z_i - Z_{i-1}} \int_{Z_{i-1}}^{Z_i} \varepsilon(z) dz \quad i = 2, 3, 4, 5 \quad (5)$$

3.3. Copolymerization kinetics

To accurately simulate real-world production scenarios and study the impact of operating variables on the properties of the polymer, such as polymer density, molecular weight, and dispersion index, it is crucial to incorporate detailed and appropriate polymerization kinetics into the modeling. It is generally believed that the Ziegler-Natta catalyst has a wide molecular weight distribution and multiple active centers. In this study, the three-site Ziegler-Natta polymerization kinetics for ethylene and 1-butene is adopted. Thirteen reactions are considered for modeling the polymerization, including chain initiation, chain growth, chain transfer, and catalyst deactivation, outlined in detailed in Table 3. Ethylene copolymerization kinetics have been well introduced by McAuley et al. [11]. Further introduction can be found in the Supporting Information, as well as the procedures of kinetic parameters estimation.

Table 3
Reaction kinetics of Z-N catalyst

Number	Reaction	Description
1	$\text{Cps}[\text{CAT}] + \text{CCAT} \rightarrow \text{P}_0$	The active site is activated by the cocatalyst
2	$\text{Cps}[\text{CAT}] + \text{BUTENE} \rightarrow \text{P}_0 + \text{BUTENE}$	The active site is activated by comonomer
3	$\text{P}_0 + \text{E2} \rightarrow \text{P}_0 [\text{E-SEG}]$	Ethylene chain initiation
4	$\text{P}_0 + \text{BUTENE} \rightarrow \text{P}_0 [\text{B-SEG}]$	Butylene chain initiation
5	$\text{P}_0 [\text{E-SEG}] + \text{E2} \rightarrow \text{P}_{n+1} [\text{E-SEG}]$	Ethylene-ethylene chain growth
6	$\text{P}_n [\text{E-SEG}] + \text{BUTENE} \rightarrow \text{P}_{n+1} [\text{B-SEG}]$	Ethylene-butene chain growth
7	$\text{P}_n [\text{B-SEG}] + \text{E2} \rightarrow \text{P}_{n+1} [\text{E-SEG}]$	Butene-ethylene chain growth
8	$\text{P}_n [\text{B-SEG}] + \text{BUTENE} \rightarrow \text{P}_{n+1} [\text{B-SEG}]$	Butene-butene chain growth
9	$\text{P}_n [\text{E-SEG}] + \text{H}_2 \rightarrow \text{D}_n + \text{P}_0$	Ethylene segment transfer to hydrogen
10	$\text{P}_n [\text{B-SEG}] + \text{H}_2 \rightarrow \text{D}_n + \text{P}_0$	Butene segment transfer to hydrogen
11	$\text{P}_n [\text{E-SEG}] \rightarrow \text{D}_n + \text{P}_0$	Ethylene spontaneous chain transfer
12	$\text{P}_n [\text{B-SEG}] \rightarrow \text{D}_n + \text{P}_0$	Butene spontaneous chain transfer
13	$\text{P}_0/\text{P}_n \rightarrow \text{Cds} + \text{D}_n$	Spontaneous deactivation of catalyst

Note: Cps[CAT] is an unactivated catalyst, CCAT is a cocatalyst, P_0 is a catalyst with active sites, BUTENE is a comonomer butene, E2 is ethylene monomer, $\text{P}_n[\text{E-SEG}]$ is ethylene segment (n is the number of segment), and $\text{P}_n[\text{B-SEG}]$ is butene segment (n is the number of segment), D_n is deactivated chain (n is the number of segment), Cds is a deactivated catalyst.

Table 4
Comparison of steady state simulation results and real plant data

	Item	Simulated value	Real plant data	Absolute error	Relative error
Reactor (FBR)		85.4	85.5	0.1	0.1%
Temperature	Pressure/MPa	2.3	2.3	0.0	0.0%
Temperature	Inlet temperature/ $^{\circ}\text{C}$	39.5	39.2	-0.3	0.8%
	Outlet temperature/ $^{\circ}\text{C}$	85.4	85.2	-0.2	0.2%
Heat exchanger (EX1)	Outlet temperature of hot stream/ $^{\circ}\text{C}$	39.5	39.0	-0.5	1.3%
	Inlet temperature of cold stream/ $^{\circ}\text{C}$	38.1	38.0	-0.1	0.3%
	Outlet temperature of cold stream/ $^{\circ}\text{C}$	44.7	44.5	-0.2	0.4%
Heat exchanger (EX2)	Inlet temperature of hot stream/ $^{\circ}\text{C}$	44.7	44.5	-0.2	0.4%
	Outlet temperature of hot stream/ $^{\circ}\text{C}$	30.0	30.3	0.3	1.0%
	Inlet temperature of cold stream/ $^{\circ}\text{C}$	28.0	28.2	0.2	0.7%
Compressor (CMPSR)	Outlet temperature	87.9	88.6	0.7	0.8%
Feed	Catalyst feed rate	7.0	7.0	0.0	0.0%
	Cocatalyst feed rate	7.0	7.0	0.0	0.0%
	Ethylene feed rate	50000.0	50000.0	0.0	0.0%
	Hydrogen feed rate	8.0	8.0	0.0	0.0%
	Butene feed rate	4450.0	4600.0	150.0	3.3%
	Isopentane feed rate	1360.0	1360.0	0.0	0.0%
Demineralized water (DW)	Flow rate	8000000.0	8000000.0	0.0	0.0%
	Inlet temperature	39.5	39.0	-0.5	1.3%
Cooling water (CW)	Flow rate	8000000.0	8000000.0	0.0	0.0%
	Inlet temperature	28.0	28.1	0.1	0.4%
Recycle gas components	Total flow rate	1714.4	1700.4	-14.0	0.8%
	Temperature	39.5	39.2	-0.3	0.8%
	Ethylene mole fraction	27.4	28.7	1.3	4.5%
	Hydrogen mole fraction	6.4	4.4	-2.0	45.5%
	Butene mole fraction	11.0	9.0	-2.0	22.2%
	Isopentane mole fraction	9.6	11.0	1.4	12.7%
	Nitrogen mole fraction	39.8	41.0	1.2	2.9%
	Ethane mole fraction	5.8	5.9	0.1	1.7%

3.4. Steady-state simulation result validation

According to the above-mentioned process and kinetics, a steady-state gas-phase polyethylene process includes three subsystems: polymerization reaction, circulating gas heat transfer and temperature regulating water is modeled *via* Aspen Plus. The material input data and energy input data of each unit in the process are obtained from an operating industrial process, ensuring the reliability and applicability of the model. The presence of two cycles in the process indeed poses challenges for achieving convergence in the simulation. In this study, the two circulating streams are designated as the tearing streams, and the actual industrial data is utilized as the initial values for these tearing streams. Multiple iterations are then performed while ensuring that the composition of the circulating gas remains within a certain range. Through these iterations, the steady-state

process converges, and accurate calculation results are obtained. The key results of the steady-state simulation are presented in Table 4, and compared with the corresponding industrial data. The data is from a facility of SINOPEC Zhenhai Refining & Chemical Company. The consistency observed between the simulated results and industrial data indicates the accuracy of the steady-state model.

The measured temperature distribution of the reactor is compared with the predicted values obtained from the simulation, as shown in Fig. 3. It clearly demonstrates that the fluidized bed reactor, modeled using the five continuous fully mixed reactors, accurately predicts the temperature distribution within the reactor. Fig. 4 presents the chain length distribution obtained using the three-site catalyst model using the method of moments [38]. The molecular weight distribution (MWD) can be obtained from the chain length distribution. The molecular weight and polymer

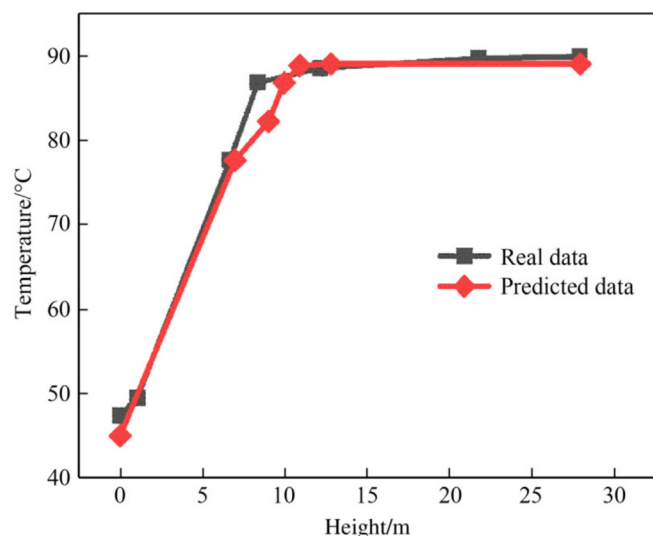


Fig. 3. Comparison of measured and predicted values of reactor temperature distribution.

dispersion index (PDI) of the polyethylene products are derived from the detailed copolymerization kinetic reactions. The high accuracy of the MWD results highlights the effectiveness of the five-cascade CSTR model in simulating the kinetic behavior of industrial fluidized bed reactors. The melt index (MI) is calculated using a correlation between MI and the weight-average molecular weight [11]. In Fig. 5, the predicted values of polymer density and MI are compared with the measured values. It can be seen from the diagram that the predicted values closely align with the industrial values for five PE grades. These findings indicate the high accuracy of the reactor modeling approach employed in this study. A sensitivity analysis of hydrogen feeding rate to melt index and polymer dispersion index (PDI), the melt index and polymer dispersion index (PDI) of the polymer increase with higher hydrogen rates. A sensitivity analysis of the C4/C2 ratio in the circulating gas on density and PDI are also conducted, both polymer density and PDI decrease as the C4/C2 ratio increases. Detailed results can be found in the Supporting Information.

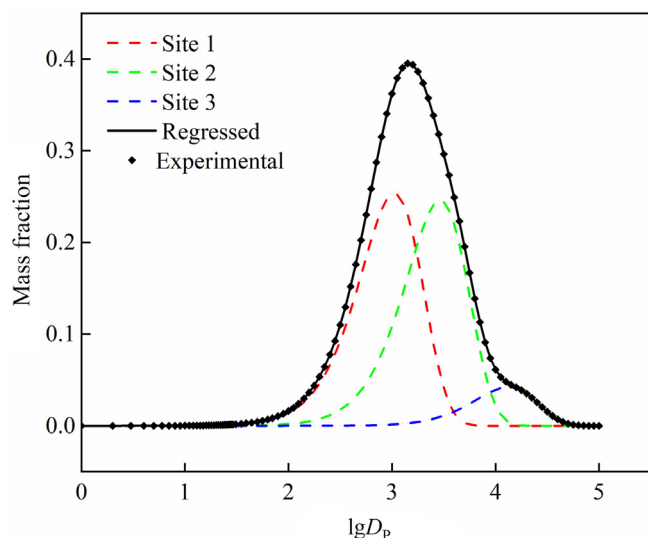


Fig. 4. The chain length distribution of the three-site model.

4. Dynamic Simulation of the Gas-phase PE Process

After the steady-state model is established and verified, the steady-state model is converted into a pressure-driven dynamic model in Aspen Dynamics. In addition, a control valve is added to control the material flow of the pipeline and to adjust the pressure. The equipment size and pressure drop of the control valve used in this paper are basically consistent with the industrial data. Notably, when there is gas flow in the model, Aspen Dynamics' pressure-driven model reflects the dynamics more realistically than the flow-driven model. We adopt the well-known integrated framework for simulation and heuristic (IFSH) method [39] to design the control scheme of the whole process. It is an insight-based heuristic method, which includes eight levels. The design method and the evaluation criteria of control schemes are introduced in the Supporting Information, while the designed control schemes for gas-phase PE plant is shown in Fig. 6.

4.1. Dynamic simulation result validation

In this study, the actuator valve of the heat exchange module in the process is specifically investigated. The valve type used is an eccentric ball valve, and its flow characteristic is depicted in Fig. 7(a). It is observed that the flow rate and valve opening exhibit highly nonlinear behavior. This nonlinearity has varying effects on heat exchange and proportional–integral–derivative (PID) control, depending on the specific valve opening range. To accurately represent this flow characteristic, the flow characteristic curve is fine-tuned based on industrial data. By incorporating this refined curve into the developed model, the relative flow rate of the valve can be simulated. In Fig. 7(b), the actual relative flow rate and the simulated relative flow rate obtained from the model are compared. The relative errors between the actual and simulated values are mostly within 10%, highlighting the accuracy between the simulated and actual values.

Two dynamic scenarios are created to validate of the model with industrial data. In the first scenario, the dynamic changes of demineralized water temperature are examined. Fig. 8(a) shows the demineralized water temperature profiles when the valve opening first increases from 41% to 63% and then decreases to 55%. In the second scenario, a reactor temperature step experiment is performed. Fig. 8(b) shows the comparison between the measured value and the simulated value when the setpoint of the reactor temperature is increased from 85 °C to 86 °C. The simulated profiles of both scenarios closely match the industrial data, confirming that the dynamic model accurately captures the real-time response of the system. Two case studies for grade transition are also conducted using the developed model, whose results can be found in the Supporting Information.

4.2. Comparative study of temperature control schemes

The reactor temperature control is the most important part of the control system, since the reactor temperature has direct effects on the production rate and product quality and more importantly the stability of the reaction system. According to the process of Fig. 1, the temperature control scheme should be related to the demineralized water temperature and/or the recycle gas temperature. In this section, we carry out eight experiments to test and compare the effectiveness of the four control schemes (namely CS1, CS2, CS3, and CS4). Conventional PID controllers are adopted in this study. Although many advanced control algorithms are developed, the reliability and operability of these algorithms should be further improved due to the model complexity and the limits of computing hardware. PID controllers still dominate the market regarding

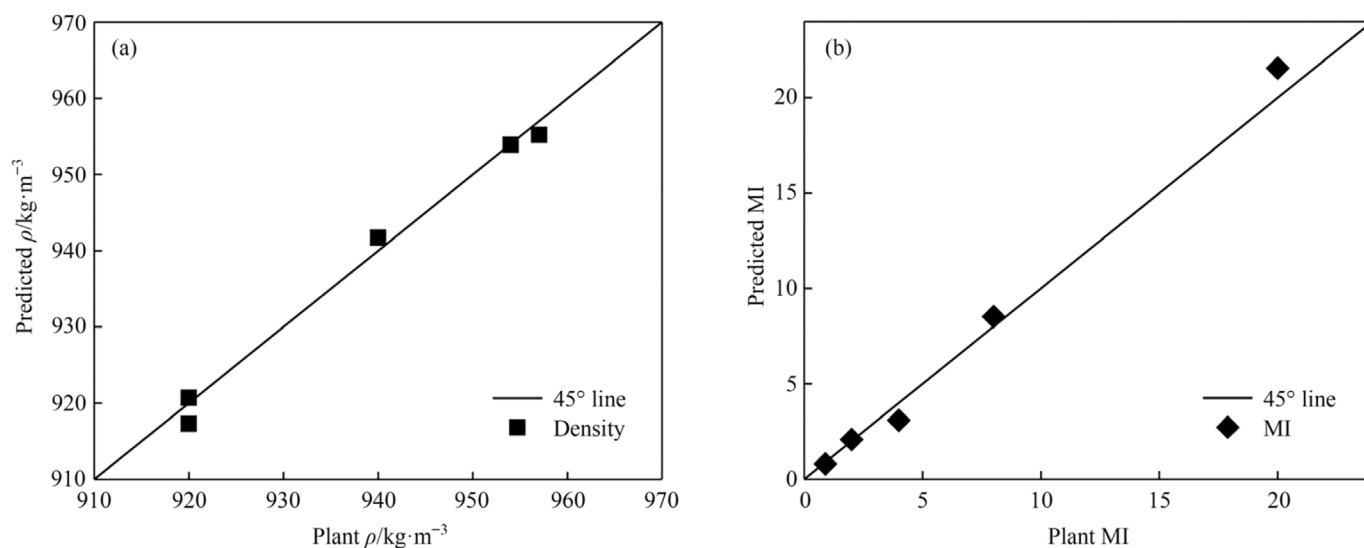


Fig. 5. Comparison between the predicted value of the model and the industrial measured value: (a) polymer density; (b) polymer melt index.

industrial gas-phase PE processes. Details of four temperature control schemes can be found in the Supplementary Material. The performance of control schemes is mainly reflected in the capability of tracking the setpoint of a key control variable and the capability of rejecting various disturbances. These two capabilities can be revealed in several aspects. Five criteria [40] are adopted, including the response speed (measured by settling time, ST), the robustness properties (measured by overshoot, OS), the capability of disturbance rejection (measured by integral absolute error, IAE),

the output smoothness (measured by total variation in the manipulated variables, TV), the economic benefits (measured by deviation from the production target, DPT). The definition of the five measuring criteria is given in the Supplementary Material.

4.2.1. Setpoint tracking and disturbance rejection capabilities of four control schemes

Experiment 1 is performed to test the setpoint tracking capability and Experiments 2–6 are performed to test the disturbances

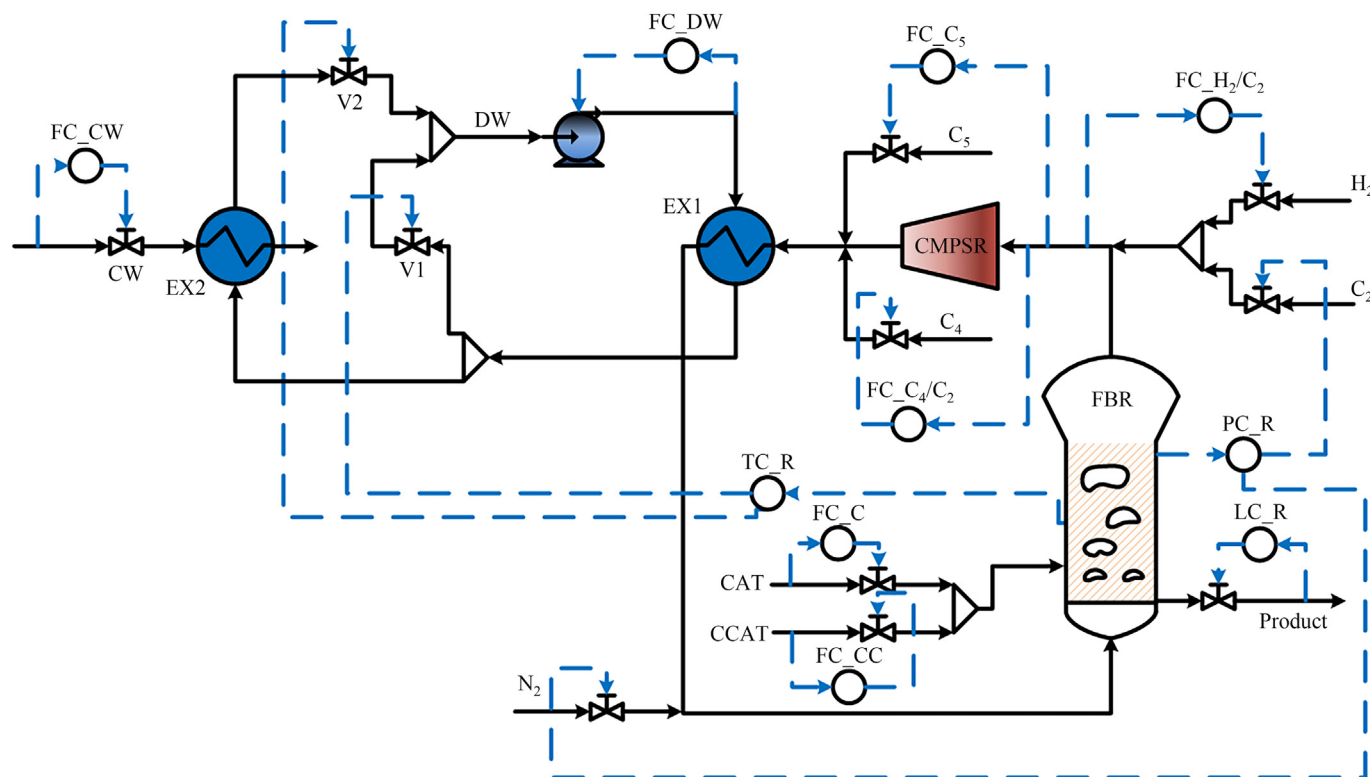


Fig. 6. Control schemes for gas-phase PE plant.

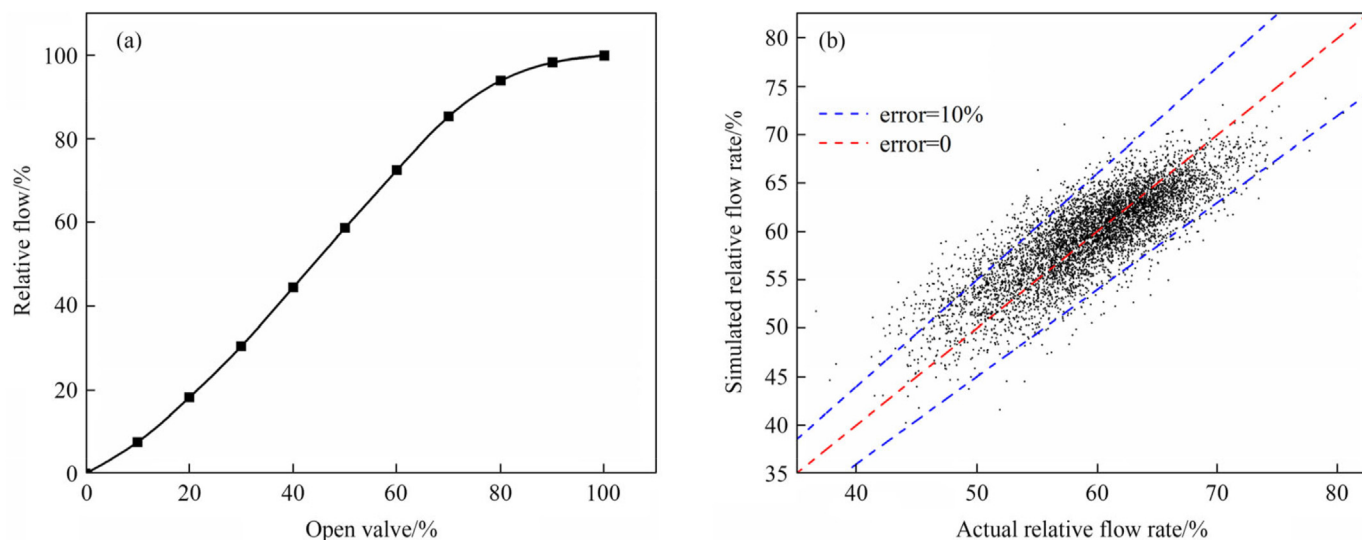


Fig. 7. (a) Valve's flow characteristic curve; (b) Comparison between actual relative flow rate and simulated relative flow rate.

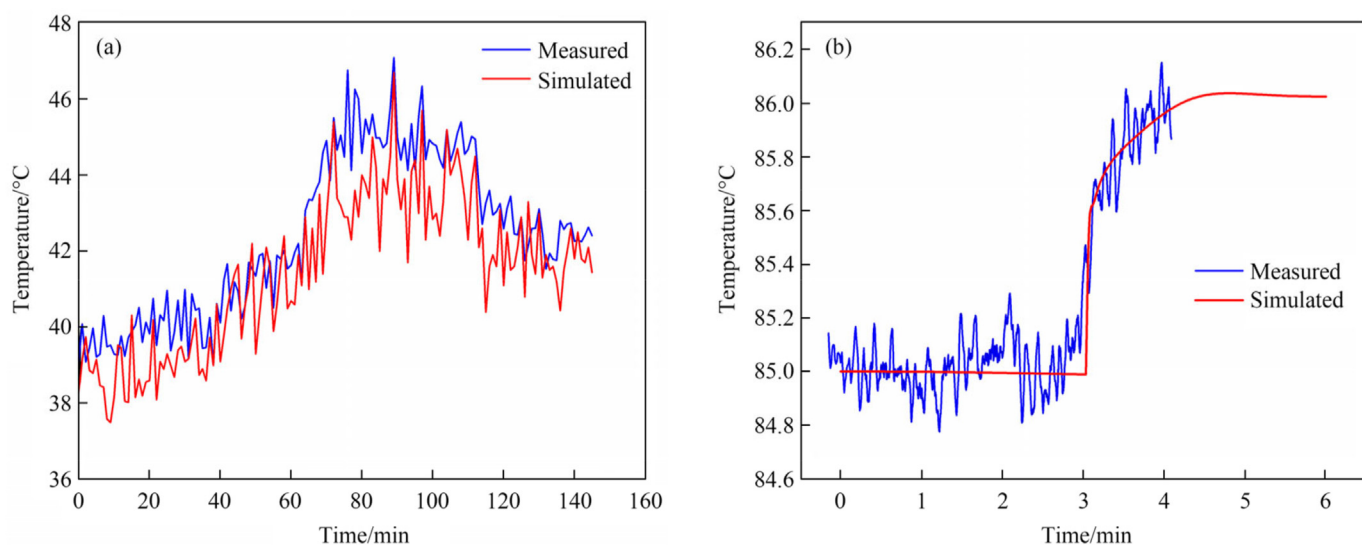


Fig. 8. Comparison between measured value and simulated value of (a) demineralized water temperature; (b) step experiment of reactor temperature.

rejection capability of four control schemes. The overall simulation time is 9 h. In the first 3 h, we keep the process running at the steady state. Then, we change the temperature setpoint or create variable disturbances at the 3rd hour moment in experiments 1–6 respectively.

The setpoint of reactor temperature is 85 °C for one of operating conditions. To examine the setpoint tracking capability of the four control schemes, the setpoint of reactor temperature is increased by 0.5 °C from 85 °C from 3 to 9 h in experiment 1 (Exp. 1). Fig. 9(a) shows the results of tracking the reactor temperature setpoint by CS1, CS2, CS3, CS4. The reactor temperature responds quickly and eventually stabilizes at the new setpoint under the both scenarios. As shown in Fig. 9(a), the four control schemes have approximately the same setpoint tracking performance. Table 5 shows the statistical results of the five performance criteria.

Grade transition is common for polymer production, while the gas components in the reactor, such as the content of comonomer butene and the concentration of chain transfer agent hydrogen, will change during the transition. It will introduce disturbances to the

reactor temperature. Heat exchange system will also bring temperature disturbances. It is important for the control scheme to have the capability of reject disturbances of the reactor temperature. The performance of the four control schemes on the reactor temperature is evaluated under the disturbances of the five variables in Experiments 2–6, including catalyst feed rate, isopentane feed rate, butene feed rate, cooling water temperature, and demineralized water flow rate. The results of Experiments 2–6 are shown in Fig. 9(b)–(e) and summarized in Table 5. Detailed analysis can be found in the Supporting Information.

4.2.2. Summary

The results of the above seven simulation experiments are summarized as follows. The best control scheme is selected for each experiment based on the criteria of ST, OS, and IAE. If the results for the two best control schemes are close, the second-best control scheme is also presented in Table 6. In terms of setpoint tracking control performance (Exp. 1), both CS2 and CS1 show good performance, while the differences between the four control schemes

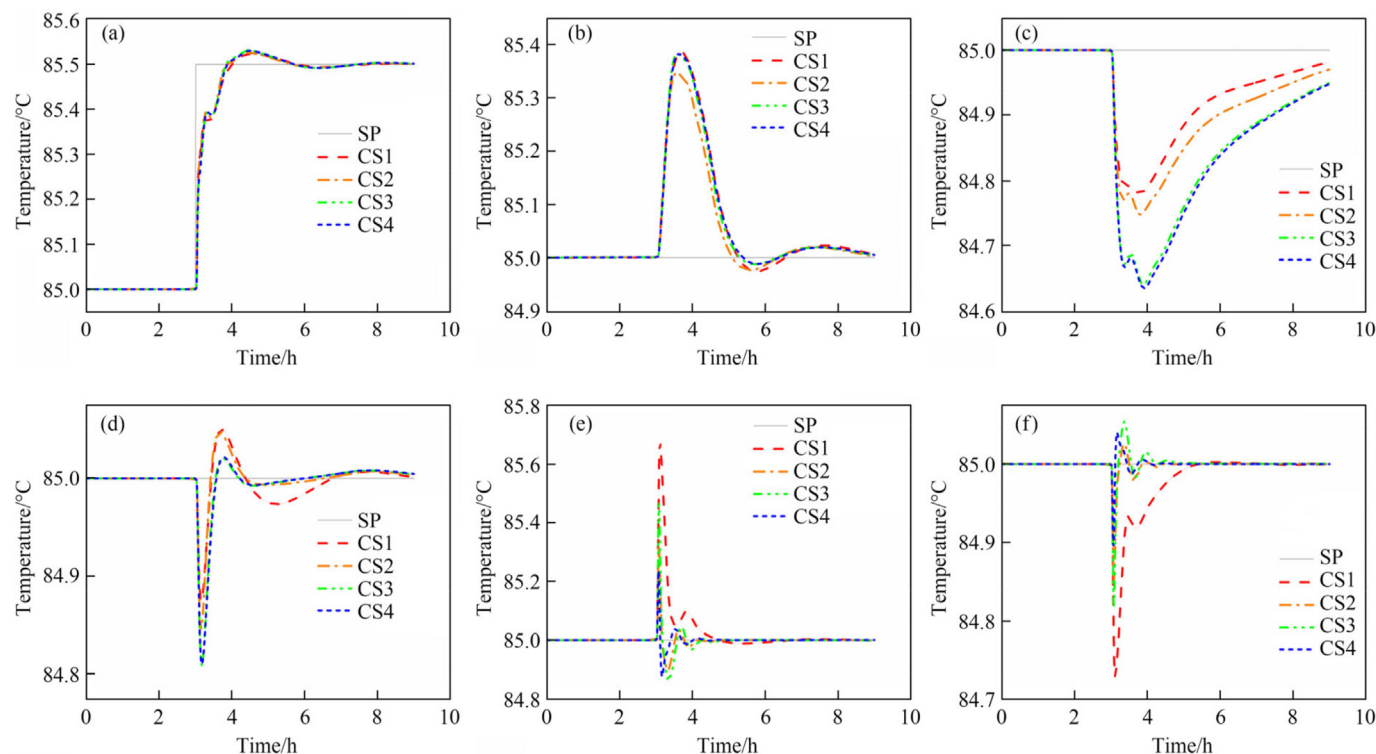


Fig. 9. Reactor temperature response to (a) Exp. 1: temperature set point increases by 0.5 °C (b) Exp. 2: catalyst feed rate increases by 10%, (c) Exp. 3: isopentane feed rate increases by 21%, (d) Exp. 4: butene feed rate increases by 8%, (e) Exp. 5: cooling water temperature increases by 1 °C, (f) Exp. 6: demineralized water flow increases by 6%.

Table 5
Performance criteria results of temperature setpoint tracking experiment (Exp. 1), catalyst feed rate fluctuation experiment (Exp. 2), isopentane feed rate fluctuation experiment (Exp. 3), butene feed rate fluctuation experiment (Exp. 4), cooling water temperature fluctuation experiment (Exp. 5), demineralized water flow fluctuation experiment (Exp. 6)

		Control scheme	performance criteria				
			ST/h	OS/°C	IAE/°C	TV/%	DPT/kg
Exp. 1	Temperature setpoint +0.5 °C	CS1	0.72	0.024	0.144	1.6567	−277772.2
		CS2	0.66	0.025	0.142	1.6782	−279173.7
		CS3	0.68	0.032	0.159	2.2408	−280114.6
		CS4	0.72	0.030	0.159	2.4488	−278966.7
Exp. 2	Catalyst feed rate +10%	CS1	1.95	0.385	0.534	4.4827	1694284.3
		CS2	1.77	0.347	0.450	4.7622	1711168.3
		CS3	1.90	0.384	0.502	4.6842	1702247.4
		CS4	1.95	0.383	0.512	4.5971	1702133.2
Exp. 3	Isopentane feed rate +21%	CS1	4.01	0.220	0.558	4.6750	−453498.9
		CS2	5.01	0.253	0.707	4.6616	−438573.0
		CS3	6.00	0.360	1.067	4.6504	−395341.6
		CS4	6.00	0.365	1.110	4.6532	−391771.9
Exp. 4	Butene feed rate +8%	CS1	0.33	0.130	0.106	0.9429	359650.8
		CS2	0.32	0.152	0.080	0.8686	386091.7
		CS3	0.41	0.180	0.083	0.7480	386909.2
		CS4	0.45	0.190	0.086	0.7251	389216.2
Exp. 5	Cooling water temperature +1 °C	CS1	1.11	0.660	0.214	1.7957	−16877.3
		CS2	0.45	0.320	0.060	2.0800	1340.9
		CS3	0.52	0.452	0.087	2.2114	417.1
		CS4	0.28	0.230	0.041	2.5345	1269.3
Exp. 6	Demineralized water flow rate +6%	CS1	1.09	0.270	0.139	1.1640	13910.3
		CS2	0.14	0.150	0.025	1.1226	3096.0
		CS3	0.16	0.180	0.035	1.1423	493.6
		CS4	0.10	0.110	0.017	1.2005	1330.4

The smallest value is highlighted in bold.

are not significant. Regarding disturbances' rejection performance, CS2 and CS1 perform well in handling disturbances related to the catalyst feed rate (Exp. 2) and the recycle gas subsystem (Exp. 3 and Exp. 4), whereas CS4 and CS2 exhibit better control performance for disturbances related to the heat exchange subsystem (Exp. 5 and

Exp. 6). Additionally, in the simulation experiment involving the coupling effect with the pressure controller (Exp. 7), CS1 and CS2 demonstrate effective control performance. In Exp. 8, where two grade transition processes are carried out, the control performance of the CS2 scheme is better than the other three schemes. Details of

Table 6
Optimal list of control schemes under different simulation experiments

Control scheme	Simulation experiments							
	Exp. 1	Exp. 2	Exp. 3	Exp. 4	Exp. 5	Exp. 6	Exp. 7	Exp. 8
CS1	✓	✓	✓	✓			✓	
CS2	✓	✓	✓	✓	✓	✓	✓	✓
CS3								
CS4					✓	✓		

Exp. 7 and Exp. 8 can be found in the Supplementary Material. Table 6 shows the optimal control schemes under different simulation experiments. It is concluded that CS2 should be selected in the temperature control of the gas-phase PE process.

5. Conclusions

Steady-state and dynamic modeling of industrial gas-phase polyethylene process, including a reaction system, a recycle gas system, and a heat exchange system, are carried out. A detailed copolymerization kinetics is established, while the kinetic parameters are obtained by fitting the MWD properties of the polymer. Considering the fluid mechanics of the fluidized bed reactor, a five-cascade CSTR model is utilized to simulate the fluidized bed reactor. The proposed model can predict the temperature distribution of the reactor accurately. The accuracy of the calculated polymer density, melt index, and molecular weight distribution is validated using real plant data. The steady-state model provides an in-depth assessment of the relationship between operating conditions and the polymer properties. This feature is important in facilitating new product development while concurrently reducing associated development costs. By adding control schemes to steady-state modules, the modeling system can be used to simulate the dynamic gas-phase polyethylene process. The accuracy of the dynamic model is also verified with real plant data. The dynamic model can be applied in both the research and the application of process control, real-time optimization, and grade transition optimization.

To demonstrate the versatility of the developed platform, temperature control of the reactor is investigated, which is recognized as a challenging task due to the highly nonlinear dynamic behavior and narrow temperature operating ranges. Eight simulation experiments are performed to evaluate four different control schemes. Since most gas-phase polyolefin processes are similar, the results of this study could provide guidelines for the control design of gas-phase polyolefin processes.

CRediT Authorship Contribution Statement

Xiaodong Hong: Conceptualization, Data curation, Methodology, Software, Writing – original draft, Writing – review & editing. Wanke Chen: Conceptualization, Data curation, Methodology, Software, Writing – original draft, Writing – review & editing. Zuwei Liao: Conceptualization, Funding acquisition, Supervision, Writing – review & editing. Xiaoqiang Fan: Conceptualization, Methodology. Jingyuan Sun: Data curation, Methodology. Yao Yang: Data curation, Methodology. Chunhui Zhao: Conceptualization, Methodology. Jingdai Wang: Funding acquisition, Supervision. Yongrong Yang: Funding acquisition, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary Material

Supplementary material to this article can be found online at <https://doi.org/10.1016/j.cjche.2024.07.026>.

Nomenclature

Ar	Archimedes number
D_p	degree of polymerization
d_p	diameter of polyethylene particle
g	gravitational constant
Re_{mf}	Reynolds number under the minimum fluidization velocity
U_{mf}	the minimum fluidization velocity
ν_f	bed porosity
ϵ_{mf}	critical bed porosity
μ	viscosity of circulating gas
ρ_g	density of circulating gas
ρ_s	density of polyethylene
ϕ	particle sphericity

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